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Social Media Use as a Predictor of Positive and Negative Affect: An Ecological Momentary Assessment Study of Adolescents with and without Clinical Depression

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Abstract

Social media use is common in adolescents, with implications for psychosocial development and the emergence of depression. Yet, little is known about the time-linked connections between social media use and adolescents' affective experiences and how they may differ between depressed and non-depressed youth. We leveraged ecological momentary assessment in adolescents oversampled for current depression to examine (1) associations between social media use and concurrent and later positive and negative affect and (2) sex and presence of a depressive disorder as moderators of these associations. Adolescents aged 14–17 with ($n = 48$) and without ($n = 97$) clinical depression, as indicated via clinical interview, reported momentary social media use and positive and negative affect seven times per day for one week. Multilevel modeling indicated that social media use was associated with reduced positive affect both concurrently and at the next assessment. Further, among clinically depressed youth only, social media use was associated with reduced negative affect at the next assessment. Results suggest that social media use may reduce *both* positive and negative affect, highlighting the nuanced relation between adolescent social media use and emotional health and laying the groundwork for future research to address several open questions.

Keywords

social media; adolescent; positive affect; negative affect; depression

Recent population-level data suggests that 62% of adolescents aged 13–18 use social media for more than two hours daily on average (Rideout et al., 2021). Consequently, social media

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CRedit Authorship Contribution Statement

Madison Politte-Corn: Conceptualization, Investigation, Data curation, Methodology, Formal analysis, Writing – original draft.

Lindsay Dickey: Methodology, Writing – review & editing, Investigation. **George Abitante:** Writing – review & editing, Validation, Resources. **Samantha Pegg:** Methodology, Investigation, Data curation, Writing – review & editing. **Christian Bean:** Methodology, Writing – review & editing. **Autumn Kujawa:** Funding acquisition, Conceptualization, Methodology, Writing – Review & Editing, Supervision.

use plays a central role in adolescents' lives, with implications for psychosocial development and the emergence of psychopathology, including depression. Some studies have found a positive correlation between social media use and depressive symptoms in adolescents, but there is variability in findings across studies (Ivie et al., 2020), and social media use may have stronger effects on short-term markers of mental health, such as positive affect (PA) and negative affect (NA; Dienlin & Johannes, 2020). Both low positive and high negative affect are commonly correlated with and shown to prospectively predict depressive symptoms in adolescents (Joiner & Lonigan, 2000; Lindahl & Archer, 2013). Yet, little is known about the temporal connections between social media use and PA and NA and whether the effects of social media use on affect differ for adolescents with and without depression.

Associations between Social Media Use and Affect

Evidence suggests that social media use has stronger effects on more immediate changes in emotionality (e.g., positive and negative affect) than longer-term or trait-like processes such as depressive symptoms or overall well-being (Dienlin & Johannes, 2020). As such, PA and NA may be key proximal processes by which social media use influences adolescent mental health yet have been largely unexamined in the existing literature. In one study of early adolescent girls, negative peer interactions that occurred on social media were more likely than in-person interactions to be associated with higher sustained, but not momentary, NA (Hamilton et al., 2021). Further, positive peer interactions that occurred on social media were more likely than in-person positive interactions to be associated with lower momentary and sustained PA (Hamilton et al., 2021). Understanding short-term relations between social media use and PA and NA may clarify how these experiences confer risk for emotional difficulties in youth.

Importantly, positive and negative affect are central correlates of depressive symptoms, such that depression is characterized by diminished PA and elevated NA (Bylsma et al., 2011; Clark & Watson, 1991; Joiner and Lonigan, 2000). Additionally, low PA and elevated NA are associated with increased risk for depression in adolescents (Abitante et al., 2022; Lindahl & Archer, 2013). Several cross-sectional studies using retrospective self-report measures have examined the association between social media use and depressive symptoms in adolescents, with some reporting a positive correlation (e.g., Ghaemi, 2020; Twenge et al., 2018) and others finding no association (e.g., Kreski et al., 2021; Schemer et al., 2020). Meta-analytic data indicates a significant but small positive correlation between social media use and depressive symptoms in adolescents ($r = .11$; Ivie et al., 2020). Further, in a nationally representative survey, teens who spent more time on social media and electronic devices were more likely to report depressive symptoms (Twenge et al., 2018). Similarly, increases in time spent on social media have been associated with increases in depressive symptoms across a 6-month period (Brunborg & Burdzovic Andreas, 2019). However, these studies do not examine temporal associations or distinguish cause from effect, and subsequent work measuring social media use and depressive symptoms at separate time points has demonstrated that social media use does not prospectively predict depressive symptoms (Beeres et al., 2021; Heffer et al., 2019). This suggests that high social media use may be a correlate of rather than a risk factor for depressive symptoms. Consistent

with this idea, other longitudinal and ecological momentary assessment (EMA) studies have found a bidirectional relation between self-esteem and frequency of social media use among women (Miljeteig & von Soest, 2022) and demonstrated that greater depressive symptoms predict more frequent social media use among adolescent girls (Heffer et al., 2019). Finally, other work highlights that the observed positive correlation between social media use and depressive symptoms may be driven specifically by problematic use, characterized by addictive tendencies (Boer et al., 2022; Cunningham et al., 2021) or through the indirect effect of co-rumination (Ohannessian et al., 2021). Taken together, this literature suggests a nuanced, and perhaps bidirectional, relation between social media use and affective health that varies as a function of psychosocial vulnerabilities and patterns of usage. Understanding when, how, and for whom social media use is associated with poor emotional functioning is paramount to clarifying the role of social media use in socioemotional development and risk for psychopathology.

Depression and Sex as Moderators of the Effect of Social Media Use on Affect

Variability in the association between social media use and mental health across studies could be due in part to moderators of this relation. This idea is consistent with the differential susceptibility to media effects model, which highlights dispositional (e.g., temperament, mood), developmental, and social susceptibility to the effects of media use on social-emotional outcomes (Piotrowski & Valkenburg, 2015). In line with this model, research suggests that adolescents with high self-esteem instability or sensitivity to peer approval benefit more from positive experiences on social media (Valkenburg et al., 2021b). This raises questions about the unique association between social media use and psychosocial outcomes among depressed youth, given that adolescent depression is associated with high self-esteem instability (Mlawer et al., 2021) and heightened sensitivity to peer feedback (Pagliaccio et al., 2022). Recent work has shown that depressed adolescents, particularly girls, are at an increased risk of problematic social media use (Shafi et al., 2021a) and demonstrate higher physiological reactivity to social media use (Shafi et al., 2021b) compared to their non-depressed peers. In addition, one recent study found that higher depressive symptoms prospectively predicted more frequent negative emotional responses to social media one year later, whereas greater positive emotional responses to social media were associated with later depressive symptoms (Nesi et al., 2022). Taken together, these findings highlight the importance of examining the ways in which adolescents' depressive symptomatology intersects with social media use to predict positive and negative emotional experiences.

In addition to depressive symptoms, past research suggests that sex and gender may moderate the association between social media use and emotional health. Specifically, girls may be more susceptible than boys to both the negative and positive effects of social media use on mental health (Frison & Eggermont, 2016; Heffer et al., 2019; Miljeteig, & von Soest, 2022). For instance, in an EMA study, recent social media use predicted lower self-esteem for women but not men (Miljeteig, & von Soest, 2022). Other work has shown that online social support negatively predicts depressed mood for adolescent

girls but not boys (Frison & Eggermont, 2016). However, not all studies have found this moderating effect of gender on the association between social media use and depression-related outcomes (Valkenburg et al., 2021b). As such, the extent to which adolescent girls are more susceptible than boys to the effects of social media use on mental health requires further investigation.

The Present Study

Taken together, the existing literature on the association between social media use and emotional health has been limited by predominantly cross-sectional designs and retrospective self-reports of average social media use, which has precluded the examination of temporal connections between social media use and concurrent and subsequent affect. Additionally, almost no research has examined the association between social media use and affect in clinically depressed youth. To address these gaps, we leveraged EMA data from adolescents oversampled for depression to (1) examine associations between social media use and concurrent and later PA and NA and (2) examine sex and current depressive disorders as moderators of these associations. We examined effects on both concurrent and later affect because cross-sectional associations could indicate that teens are more likely to use social media depending on their affect, but lagged associations allow us to examine whether social media use predicts affect at later time points. We hypothesized that social media use would predict lower concurrent and later PA and higher concurrent and later NA. Further, we expected that these associations would be stronger for female compared to male adolescents and for depressed versus non-depressed youth. This study was not pre-registered.

Method

Participants

Participants were 156 adolescents aged 14–17 ($M = 15.23$; $SD = 1.08$) at varying risk for depression recruited as part of a study on social and emotional processing. Eleven participants either did not complete any EMA surveys ($n = 9$) or completed surveys that did not pass our data quality checks described below ($n = 2$) and were not included in the current analyses. Of the remaining 145 participants, 97 had no history of clinical depression and 48 met diagnostic criteria for a current depressive disorder. Of these currently depressed youth, 16.3% met criteria for joint major depressive disorder (MDD) and persistent depressive disorder (PDD), 32.7% met for MDD only, 20.4% met for PDD with intermittent major depressive episodes (MDE) including a current MDE, 22.4% met for PDD with intermittent MDE without current MDE, 6.1% met for PDD without a history of MDE, and one adolescent met for unspecified depression. Demographic data were acquired via self- and parent-report on a questionnaire. Regarding gender distribution in the full sample, 90 participants (62.1%) self-identified as female, 53 (36.6%) identified as male, and 2 (1.4%) identified as nonbinary. As such, we were underpowered to examine associations between the study variables and each gender group and focus subsequent analyses on biological sex based on a combination of parent and adolescent reports (female $n = 93$; male $n = 52$). The majority of participants identified as White ($n = 104$; 71.7%), 19 (13.1%) identified as

Black or African American, 10 (6.9%) identified as Asian, 1 (0.7%) identified as American Indian or Alaska Native, 2 (1.4%) identified as Native Hawaiian or Pacific Islander, and 9 (6.2%) identified as another race. Nine participants (6.2%) further identified as Hispanic or Latinx. To assess socioeconomic status, mothers reported on education level and annual income for themselves and a second parent, if applicable. Of the highest education level across both parents, 3 (2.1%) were high school graduates, 9 (6.2%) attended college or trade school but did not finish, 14 (9.7%) graduated a 2-year college or trade school, 45 (31.0%) graduated a 4-year college, 46 (31.7%) earned a Master's degree, 26 (17.9%) earned a doctoral degree (e.g., MD, PhD, JD), and 2 (1.4%) did not report on education level. Of combined annual income across both parents, five (3.4%) reported income below \$30,000, 37 (25.5%) between \$30,000-\$89,999, 53 (36.6%) between \$90,000-\$149,999, 28 (19.3%) between \$150,000-\$199,999, 20 (13.8%) reported an annual income of \$200,000 or higher, and 2 (1.4%) did not report on annual income.

Procedure

Adolescents were recruited via a university medical center listserv, flyers distributed to local pediatric and mental health clinics, Facebook ads, and word of mouth. Exclusion criteria were use of antipsychotics or mood stabilizers, developmental delays, intellectual disabilities, visual or hearing impairments, history of psychosis or bipolar disorder, and history of depression without a current depression diagnosis. Inclusion criteria were presence of a current depressive disorder or no history of clinical depression. In total, 231 adolescents were deemed eligible for the study based on an initial phone screen with their biological mothers, conducted by research staff members or graduate students. Of these, 156 (67.5%) completed a clinical interview to confirm the presence or absence of a depressive disorder. Parents of adolescents provided informed consent, and all adolescents provided informed assent prior to their clinical interview. The EMA period began the day after participants completed their diagnostic interview, which could occur on any day of the week and at any time of year. The average length of participation from the initial phone screen conducted with adolescents' mothers to the end of the EMA period was 56 days. In total, 147 of these adolescents provided EMA data. The Institutional Review Board at Vanderbilt University approved this study.

Measures

Diagnostic Interviews—Diagnostic interviews were conducted prior to the EMA period. To assess current and lifetime diagnoses of depression, the DSM-5 version of the Schedule for Affective Disorders and Schizophrenia for School-Aged Children 6–18 years (K-SADS; Kaufman et al., 2013) was administered by clinical psychology doctoral students or masters-level clinicians. Interviews were supervised and diagnoses were verified by a licensed clinical psychologist (Dr. Kujawa). The average age of onset for current depressive episodes was 14.17 years ($SD = 1.85$) for MDD and 12.79 years ($SD = 2.27$) for PDD, with an average episode duration of 63.76 weeks ($SD = 87.37$; range = 3–364) for MDD and 137.39 weeks ($SD = 108.17$; range = 52–468) for PDD. To evaluate inter-rater reliability, a subset of audiotaped interviews was reviewed and coded by an independent interviewer with excellent inter-rater reliability for depressive disorders ($kappa = 1.00$).

Ecological Momentary Assessment—Measures of social media use, PA and NA were obtained through EMA. Participants received text messages through SurveySignal seven times per day for one week (SurveySignal; Hofmann & Patel, 2015). Surveys were sent on a variable schedule and participants were given two hours to respond.¹ Each survey prompted participants to report what they were doing in the moment including using social media, who they were with both in person and online, and current PA and NA.

The mean number of assessments completed was 27.30 ($SD = 12.72$); in total, we obtained 4,052 assessments nested within 147 participants. Sixteen surveys were completed outside of the EMA period and were not included in the analyses. As additional quality control checks for the EMA data, we excluded surveys completed in under 30 seconds ($n = 39$) or over 15 minutes ($n = 63$) and surveys completed within 15 minutes of the prior assessment ($n = 24$) resulting in a final sample of 3,910 assessments nested within 145 participants. To test the robustness of our results and mitigate concerns about potential bias due to responses not being missing at random (Lafit et al., 2023; Trull & Ebner-Priemer, 2020), all main analyses were re-run excluding participants who completed 20% or fewer of all possible assessments ($n = 22$) and again excluding participants who completed 50% or fewer of all possible assessments ($n = 53$). Applying these compliance thresholds did not change the significance or direction of observed effects (see Supplemental Information). Therefore, results including the entire sample are reported.

We also assessed whether EMA compliance was impacted by survey prompts being sent during the school day. A paired samples t-test indicated that there were no significant differences in survey completion on weekend days (i.e., Saturday and Sunday; $M = 54\%$ of surveys completed) versus weekdays ($M = 56\%$ of surveys completed), $t(144) = -1.55$, $p = .12$, Cohen's $d = -.09$. Further, the average number of surveys completed did not significantly differ for participants who started the EMA period from August to May ($M = 33.32$), which is generally consistent with the academic calendar for area public schools, compared to individuals who began the EMA period in June or July ($M = 32.68$), $t(144) = 1.64$, $p = .11$, Cohen's $d = .07$.

Social Media Use.: On each survey, participants were asked “*What were you doing when you received this survey? Please select all that apply*” and presented with a checklist of several possible activities, the first of which was “*Browsing or posting on social media.*” As such, social media use was coded as either 0 (not browsing or posting on social media) or 1 (browsing or posting on social media) for each survey. Across all surveys included in the analyses, 480 (12.3%) included an endorsement of social media use. Of these observations, 147 (30.6%) were completed by depressed youth and 350 (72.9%) were completed by female adolescents.

Positive and Negative Affect.: Affect was measured on each survey using a brief version of the Positive and Negative Affect Scales for Children (PANAS-C; Ebesutani et al., 2012; Laurent et al., 1999). Participants were asked to indicate the extent to which they felt each

¹Fifty participants received surveys between 6:25am and 10pm. Due to a change in software, 95 participants were set on a different time zone and received surveys between 8:25am-12am, and we covaried for this time difference in analyses.

emotion at that moment. At each assessment, PA and NA were calculated by averaging responses to the 10 items which were rated on a Likert-type scale from 1 (very slightly or not at all) to 5 (extremely).² PA was calculated by averaging five items (joyful, proud, cheerful, happy, and lively), and NA was measured by another five items (miserable, mad, scared, afraid, and sad). The PANAS scales display high internal consistency (alphas ranging from .89 - .94) and strong convergent and discriminant validity with other measures of mood in samples of children and adolescents (Laurent et al., 1999) and the brief NA and PA scales display equivalent psychometric properties (Ebesutani et al., 2012). Across all surveys, Cronbach's alpha indicated that overall internal consistency in the current sample was high for both PA ($\alpha = .92$) and NA ($\alpha = .80$). We also examined whether internal reliability changed across the EMA period. There was a pattern indicating that internal consistency for PA was relatively higher after the first day of the EMA period ($\alpha = .88$ on day one and $\alpha = .92 - .93$ on days two through seven). Internal consistency for NA was relatively higher during the final days of the EMA period ($\alpha = .74 - .79$ on days one through five and $\alpha = .84 - .86$ on days six and seven).

Data Analysis—To account for the hierarchical structure of our data, multilevel modeling (also known as hierarchical linear modeling) was used to examine the effects of social media use on PA and NA. Multilevel modeling is also advantageous for analyzing EMA data because it does not assume that data points are independent and can handle missing values (Snijders & Bosker, 2011). The data were hierarchically arranged in three levels, with assessments (level 1) nested within days (level 2) nested within participants (level 3). Level-1 variables (measured at each assessment) included PA, NA, and social media use. We did not include any level-2 variables (i.e., day of the week) in the models, as the fixed effects of weekday on PA and NA were nonsignificant and did not improve model fit ($\Delta AIC = +1.38$, $\Delta BIC = +7.64$ for PA; $\Delta AIC = -0.82$, $\Delta BIC = +5.45$ for NA; log-likelihood test p 's $> .05$). Additionally, the pattern of results did not change when including weekday as a level-2 covariate (see Supplemental Information). Level-3 variables (i.e., person-level variables) included sex and presence/absence of depressive disorder.

We used an iterative model build-up strategy to test the interactive effects of social media use with sex and depression on concurrent PA and NA. All analyses were conducted in *R* using the *nlme* package (Pinheiro et al., 2020). Maximum likelihood was used as the estimation method. For each outcome variable (i.e., PA and NA), we began with unconditional (or null) models and then imposed a lag-1 continuous autoregressive correlation structure on the residuals. Subsequently, we introduced predictors as fixed effects and examined the slope variance of level-1 predictors to determine whether they should have random slopes. Finally, we introduced cross-level interactions between social media use and sex and depression group and probed significant interactions using simple slopes. We estimated Cohen's d for all parameter estimates using the *EMATools* package in *R*, though effect sizes for multilevel models should be interpreted with caution. All participants were included in the analysis regardless of the number of assessments completed, as maximum likelihood estimation accommodates missing data as part of the estimation process and the

²Using summed scores of PA and NA did not change the significance or direction of observed effects.

pattern of results was unchanged when excluding participants with low completion rates (less than 20% and 50%; Peugh & Enders, 2004).

Results

Descriptive Statistics

Means, standard deviations, and correlations between study variables are presented in Table 1. Across assessments, PA and NA were moderately negatively correlated. As expected, non-depressed teens reported higher average PA, $t(141) = 4.54, p < .001$, Cohen's $d = -.81$, and lower average NA, $t(141) = -4.03, p < .001$, Cohen's $d = .87$, than depressed teens. Independent samples t-tests indicated that female adolescents reported lower average PA than male adolescents, $t(141) = 3.079, p = .002$, Cohen's $d = .55$, but there were no sex differences in average NA, $t(141) = -1.34, p = .18$, Cohen's $d = -.22$, or frequency of social media use (the proportion of surveys in which social media use was endorsed³; $t(136) = -.63, p = .53$, Cohen's $d = -.12$). A chi-square test with Yates' continuity correction indicated that sex was not significantly associated with depression diagnosis, $\chi^2(1) = 3.01, p = .08$. There was a pattern for a higher proportion of female adolescents in the depressed group than male adolescents (38% of females versus 23% of males), consistent with well-established sex differences in depression (Allgood-Merton et al., 1990; Hankin et al., 2007). Frequency of social media use was not significantly associated with average PA or NA, nor were there differences in social media use between depressed and non-depressed adolescents, $t(136) = 1.03, p = .15$, Cohen's $d = -.19$. Thirty-seven participants reported no social media use across surveys, and these participants did not differ from teens endorsing any social media use on average PA or NA or depression status ($ps > .05$). Age and the number of assessments completed were not significantly correlated with any of the other study variables, and the pattern of results was unchanged when including the EMA start month (June or July versus school year months) as a level-3 covariate (see Supplemental Information). As such, we report the results of the models without these covariates for parsimony.

Unconditional Models

We began with two unconditional (null or random-intercepts only) models with PA or NA as the dependent variable to compute the intraclass correlation (ICC) and partition the variance in affect across levels. Consistent with the recommendation by Hawkey et al. (2007), we computed each model with three-levels given that assessments were nested within days, and days were nested within participants. The ICCs for PA and NA supported the use of a three-level model, as there was a sizable amount of variance in affect that occurred across assessments, across days, and between subjects (37%, 8%, and 55% for PA; 37%, 10%, and 54% for NA).

The Effect of Social Media Use on Concurrent Momentary PA and NA

Using the null models for PA and NA as baselines (AIC = 8043.66, BIC = 8068.71 for PA; AIC = 3224.37, BIC = 3249.42 for NA), we proceeded with model build-up. We first

³Participants who completed <5 assessments ($n = 8$) were not included in analyses with this variable.

tested if imposing a lag-1 continuous auto-regressive correlation structure on the residuals improved model fit. As expected, given time dependencies across assessments, model fit was improved ($\Delta AIC = -30.21$, $\Delta BIC = -23.95$ for PA; $\Delta AIC = -36.54$, $\Delta BIC = -30.28$ for NA). Consequently, we retained a lag-1 continuous auto-regressive correlation structure in all subsequent models.

Results of analyses examining concurrent momentary PA and NA (i.e., PA and NA reported at the same assessment) are presented in Table 2. Consistent with our first goal, we next tested the fixed concurrent effect of social media use on PA and NA. Social media use was associated with lower concurrent PA ($\beta = -.04$, $SE = .01$, $p < .001$, Cohen's $d = -.14$), and the inclusion of this predictor improved model fit ($\Delta AIC = -16.65$, $\Delta BIC = -4.12$). Social media use was not associated with concurrent momentary NA ($\beta = -.01$, $SE = .01$, $p = .36$, Cohen's $d = -.06$), nor did model fit improve with the addition of this predictor ($\Delta AIC = -5.20$, $\Delta BIC = +1.06$). Nevertheless, we retained social media use as a predictor in both models in order to examine potential moderating effects.

Next, we examined whether social media use should have a random slope, indicating that the concurrent effect of social media use on PA or NA varies across days and individuals. We examined the slope variance and within-person range of slopes to determine whether a random slope was warranted. The slope variance was significant for both PA ($\tau = .041$; $p = .04$) and NA ($\tau = .016$; $p = .02$). Specifically, the between-person effect of social media use on PA varied from $\beta = -.24$ to $\beta = .16$, with all but one participant showing a negative association between social media use and PA. The between-person effect of social media use on NA varied from $\beta = -.21$ to $\beta = .19$, with the majority of participants (70.6%) showing a negative association between social media use and NA. Thus, we retained a random slope for social media use in both cross-sectional models.

Moderators of the Concurrent Effect of Social Media Use on Momentary PA and NA

Next, we examined moderators of the concurrent effect of social media use on momentary PA and NA. We first added the potential time-invariant moderators as level-3 predictors, covarying for concurrent social media use at level 1, to examine any main effects. Presence of a depressive disorder was a significant predictor of both PA and NA ($\beta = -.24$, $SE = .06$, $p < .001$, Cohen's $d = -.70$ for PA; $\beta = .29$, $SE = .06$, $p < .001$, Cohen's $d = .81$ for NA). There was also a significant main effect of sex on PA ($\beta = -.15$, $SE = .06$, $p = .01$, Cohen's $d = -.44$) but not NA ($\beta = .03$, $SE = .06$, $p = .66$, Cohen's $d = .07$), indicating that female adolescents reported lower PA than male adolescents. Nevertheless, sex was retained as a predictor in both models to examine moderating effects, consistent with the study aims.

Finally, we added cross-level interaction terms using social media use (level 1) and person-level (level 3) variables to examine moderating effects, retaining a random slope for social media use (see Table 2). Neither teen depressive disorder nor sex significantly moderated the concurrent effect of social media use on PA or NA ($ps > .05$).

The Effect of Social Media Use on Later PA and NA

We next repeated the model-build up process using the lagged effect of social media use on PA and NA, allowing us to make stronger claims about temporal associations. The average

time between completed assessments, including overnight lags, was 5.32 hours ($SD = 6.41$ hours), with lags ranging from 15 minutes to 4 days after the prior assessment. Excluding overnight lags, the mean time between assessments was 2.90 hours ($SD = 1.64$ hours), with lags ranging from 15 minutes to 17.22 hours after the prior assessment. To minimize variability in the duration between assessments, and because we expected the prospective association between social media use and affect to emerge for shorter rather than longer time intervals, we excluded overnight lags for subsequent analyses (updated $n = 2,953$ assessments nested within 138 participants).

Results of the regression analyses predicting affect at the next completed assessment are reported in Table 3. We began with null models for PA and NA with a lag-1 continuous auto-regressive correlation structure (AIC = 6145.08, BIC = 6175.03 for PA; AIC = 2454.09, BIC = 2484.04 for NA). Next, we tested the fixed lagged effect of social media use on PA and NA at the next assessment. Social media use was associated with lower PA at the next assessment ($\beta = -.03$, $SE = .01$, $p = .009$, Cohen's $d = -.11$), although the inclusion of this predictor did not substantially improve model fit ($\Delta AIC = -7.71$, $\Delta BIC = +4.28$). This effect remained significant when controlling for PA at the prior assessment and duration between assessments ($\beta = -.03$, $SE = .01$, $p = .02$). Social media use was not associated with NA at the next assessment ($\beta = -.02$, $SE = .01$, $p = .14$, Cohen's $d = -.04$), nor did model fit substantially improve with the addition of this predictor ($\Delta AIC = -5.18$, $\Delta BIC = -.82$). However, we retained social media use as a predictor in subsequent models to examine moderating effects.

We next examined whether social media use should have a random slope, indicating that the lagged effect of social media use on PA or NA varies across individuals. The slope variance for the lagged effect of social media use was not significant for PA or NA ($\tau = .015$; $p = .35$ for PA; $\tau = .004$; $p = .46$ for NA). Thus, we reverted to a fixed slope for social media use in both models predicting later affect.

Finally, given evidence that affect may influence the likelihood of engaging with social media (Heffer et al., 2019; Miljeteig & von Soest, 2022), we examined whether affect predicted social media use at the next completed assessment in order to make stronger directional claims, again excluding overnight lags and reverting to a fixed slope. Neither PA nor NA was significantly associated with social media use at the next assessment ($B = .13$, $SE = .09$, $p = .13$ for PA; $B = .03$, $SE = .15$, $p = .86$ for NA).

Moderators of the Lagged Effect of Social Media Use on Momentary PA and NA

We next proceeded to test interactive effects between social media use and sex and depressive disorder on later PA and NA. The main effects of gender and teen depressive disorder on PA and NA were not substantively changed when covarying for lagged rather than concurrent social media use. As seen in Figure 1, we found a significant interaction between lagged social media use and presence of a depressive disorder on later NA, $\beta = -.04$, $t(132) = -2.39$, $p = .02$, Cohen's $d = -.10$. Simple slopes analyses revealed that social media use was not associated with later NA for adolescents with no history of clinical depression [simple slope = .005, $SE = .03$, $t = 0.18$, $p = .86$]. However, social media use was associated with lower NA at the next assessment for currently depressed adolescents [simple

slope = $-.10$, $SE = .04$, $t = -2.81$, $p = .006$]. It is important to note that most adolescents in the current sample, particularly those with no history of depression, reported very little or no NA (among never-depressed youth, 64% of NA ratings were at the lowest possible value, compared to 37% of ratings in the depressed group). Consequently, the observed negative association between social media use and NA in depressed teens only may be due to floor effects. This pattern of results was not substantively changed when covarying for NA at the prior assessment or duration between assessments. Teen depressive disorder did not moderate the lagged effect of social media use on PA, nor were there any interactions between lagged social media use and sex on later PA or NA ($ps > .10$). Finally, given heterogeneity in depression diagnoses in the current sample, we ran supplementary analyses within the depressed group to examine the moderating effect of depression diagnosis (i.e., MDD vs. PDD) on the association between social media use and affect (see Supplemental Information).

Discussion

The aims of the present EMA study were to (1) examine associations between social media use and concurrent and later PA and NA and (2) examine sex and depressive disorder as moderators of these associations in a sample of adolescents. Consistent with hypotheses, social media use was associated with reduced PA both concurrently and at the next assessment. Unexpectedly, social media was also associated with reduced NA at the next assessment, but only for currently depressed youth. Inconsistent with study hypotheses, sex did not moderate the association between social media use and PA or NA, though we did find that female adolescents reported significantly lower average PA than males. To our knowledge, this study was among the first to examine the time-linked connections between adolescent social media use and affect and how this association may differ for clinically depressed youth. Results suggest that use of social media may reduce both positive and negative affect, highlighting the nuanced relation between social media use and mental health outcomes in adolescents.

Effect of Social Media Use on PA

Our finding that social media use was associated with lower PA is consistent with a prior EMA study, which demonstrated that peer interactions on social media were associated with lower levels of PA at the following assessment (Hamilton et al., 2021). Moreover, this finding is generally consistent with the broader literature showing a positive correlation between social media use and depressive symptoms in adolescents (Ghaemi, 2020; Ivie et al., 2020), given that low PA is associated with depressive symptoms and risk (Bylsma et al., 2011; Lindahl & Archer, 2013). However, it is important to note that we did not find associations between overall social media use and affect, providing empirical support for claims that social media use may have stronger effects on short-term markers of emotional health than more stable measures (Dienlin & Johannes, 2020). This may also help to explain mixed findings and small effects across the literature on social media use and affective psychopathology, which has largely relied on measures of overall mental health rather than examining short-term fluctuations in affect (e.g., Ghaemi, 2020; Ivie et al., 2020; Kreski et al., 2021; Schemer et al., 2020). At the between-person level, we demonstrated that all but

one participant showed a negative association between social media use and PA, compared to only 56% of participants showing a negative association between social media use and NA. This suggests a more global association between social media use and reduced PA in adolescents, whereas the association between social media use and NA is variable. Prior work has highlighted a number of variables that might explain this dampening effect on PA, such as passive rather than active use of social media (Dienlin & Johannes, 2020), social comparison (Nesi & Prinstein, 2015), and co-rumination (Ohannessian et al., 2021). Further, past research suggests that even positive peer interactions on social media do not confer the same emotional benefits as in-person positive peer interactions (Hamilton et al., 2021).

Our findings add to the predominantly cross-sectional body of literature on the relation between social media use and depressive symptoms by leveraging EMA to examine temporal associations. Specifically, cross-sectional associations in prior studies have raised the question of whether social media use is an indicator of emotional difficulties rather than a risk factor. We demonstrated that social media use prospectively predicted reduced PA at the subsequent assessment, but PA did not predict subsequent social media use, suggesting that social media use is a risk factor for negative emotional experiences in adolescents. Further, this effect remained significant when covarying for PA at the prior observation, which demonstrates that the observed effect reflects reductions in PA across time rather than a sustained dampening effect on PA.

Effect of Social Media Use on NA

Unexpectedly, we did not find that social media use was associated with greater NA in the overall sample. This may suggest that the harmful effects of social media use on affect result from reducing PA rather than increasing NA. Interestingly, this is consistent with growing evidence from neuroscience that reduced activation of positive valence systems is a unique and key predictor of depression risk (Kujawa & Burkhouse, 2017; Kujawa et al., 2020). Alternatively, the effects of social media use on NA may be more variable and depend on the type of use. We did find an effect of social media use on NA in the opposite direction only for currently depressed youth, such that social media use was associated with reduced NA both concurrently and at the next assessment. This could mean that depressed adolescents use social media to distract from negative emotions, but that social media use may blunt emotions more generally rather than only blunting NA. It is important to note that most adolescents in the current sample, particularly those with no history of depression, reported very little or no NA (among never-depressed youth, 64% of NA ratings were at the lowest possible value, compared to 37% in the depressed group). Consequently, it is possible that we were unable to detect associations with NA in the full sample due to floor effects.

Our finding that social media use reduced NA for currently depressed youth may be explained in part by the social compensation hypothesis, which posits that youth with low or unstable self-esteem experience increases in self-esteem following use of social media, which compensates for the lack of positive experiences in their offline lives (Valkenburg & Peter, 2011). In line with this reasoning, other work has found that greater positive emotional responses to social media prospectively predicted depressive symptoms one year later (Nesi et al., 2022). Taken together, it could be the case that some of the characteristics

that may make social media more salient and enjoyable, such as sensitivity to peer approval, self-esteem instability, and lack of positive in-person experiences, may also overlap with risk factors for the development of depressive symptoms. It is possible that although social media use may reduce NA in the short term for some adolescents, it may still be associated with depressive symptoms longer term. Alternatively, another growing line of evidence demonstrates that online social support, the provision of emotional support and social companionship via online means, negatively predicts depressive symptoms (Cole et al., 2017; Nick et al., 2018), and this effect appears to be particularly strong for adolescents (Politte-Corn et al., 2023). This suggests that the nature of social media experiences and adolescents' perceptions of online social support may be key in determining which teens experience the potential benefits of social media use on emotional health. Delineating how and for whom social media use may confer mental health benefits and whether these beneficial effects persist longitudinally is a key direction for future research.

Sex as a Moderator of the Relation between Social Media Use and Affect

Finally, we did not find moderating effects of sex on the association between social media use and affect, inconsistent with study hypotheses. There is mixed evidence regarding sex differences in the effect of social media use on mental health outcomes (e.g., Frison & Eggermont, 2016; Valkenburg et al., 2021b). One possibility for these mixed findings is that specific types of online experiences may differentially impact girls and boys, but these are not captured by generic measures of social media use that have dominated existing literature. For instance, female adolescents may be more susceptible than male adolescents to positive and negative peer interactions on social media (Frison & Eggermont, 2016; Hamilton et al., 2021). Another possibility is that we were underpowered to detect moderation by sex, given that only 27% of surveys with an endorsement of social media use were completed by males (132 surveys total; 3.3% of assessments collected).

Limitations and Future Directions

The present study has several strengths, including the use of EMA to study social media use and affect in real-time, three-level hierarchical linear models to leverage all data and minimize standard errors, a sample of adolescents with and without clinical depression, and longitudinal data which allowed us to make stronger inferences about causality. However, there are also limitations that could guide future work. First, we obtained a relatively low response rate, perhaps because our incentive structure was not strong enough or because many teens are unable to complete surveys during the school day. Although sensitivity analyses with varying compliance rates support the robustness of our effects, we did not assess reasons for missingness and so cannot rule out the possibility that missed responses were influenced by an unmeasured variable. In particular, common reasons for missed responses in EMA studies with adolescents such as tendencies to sleep in late or amount of time spent studying or completing homework should be assessed in future studies (Janssen et al., 2021). Second, many of our adolescents, particularly those without current depression, reported very little or no NA, which may have hindered our ability to detect stronger effects of social media use on dampened NA. To disentangle whether currently depressed youth are uniquely susceptible to the potentially beneficial effects of social media use on short-term affective health, future work could use samples of youth with elevated NA in the

absence of current clinical depression (e.g., youth at temperamental risk for internalizing psychopathology or anxious youth). Relatedly, we observed high heterogeneity in the effect of social media use on NA, such that between-person slopes varied from $\beta = -.21$ to $\beta = .19$, with nearly a third of our sample showing a positive association between social media use and NA. This variable association between social media use and NA, and whether effects on PA versus NA more strongly impact emotional health and psychopathology risk, requires further investigation.

Third, our measure of social media use was limited, and we did not include any level-1 predictors to explain within-person variance in the effect of social media use on affect. Specifically, we did not define ‘social media’ for our participants, so it is possible that there was confusion or misperception of what should be included in this category. Moreover, specific types of social media interactions or experiences may have unique effects on PA and NA, but this was not captured by our single-item measure. An important direction for future research is to identify the proximal social media experiences and contexts which moderate the association between social media use and affect within and between individuals. Relatedly, our measure of social media use did not differentiate active versus passive use, which may be an important distinction impacting affective outcomes (Clark et al., 2018; Dienlin & Johannes, 2020). Specifically, active use of social media (e.g., liking and commenting on others’ posts, interacting with others online) may facilitate social connection and beneficially impact affective change, while passive use such as scrolling and watching others’ content without social interaction may be more detrimental to affect (Clark et al., 2018). Fourth, the current study tested biological sex as a moderator of the effects of social media use on PA and NA to compare to prior work, but gender or sexual minority status may be a more meaningful moderator. For instance, a recent study demonstrated that sexual minority adolescents display distinct behavioral and neural reactions to peer feedback (Clark et al., 2023), suggesting that social media use may have unique effects on the emotional well-being of sexual and gender minority youth. Additionally, there was a higher proportion of female adolescents in the depressed group than male adolescents. While this is consistent with well-established sex differences in adolescent depression (Allgood-Merton et al., 1990; Hankin et al., 2007), this may have hindered our ability to obtain fair estimates of these two moderating effects. Finally, our sample was predominantly White. Other research has shown distinct processes by which social media use increases risk for internalizing symptoms for adolescents of color (Tao & Fisher, 2022; Thomas et al., 2023). As such, the time-linked connections between social media use and affect among ethnic and racial minority adolescents may differ from those observed in the present study.

Implications

These findings may have implications for prevention and intervention of affective psychopathology in adolescents. First, adolescents experiencing blunted positive affect may be encouraged to decrease their overall usage of social media. At the same time, given our finding that social media use was associated with reduced NA at the next assessment for clinically depressed youth, solely discouraging social media use for depressed adolescents should be avoided. Instead, parents and clinicians could encourage limited but healthy, active usage of social media that facilitates social connection, given other work indicating

that online social support is associated with lower depressive symptoms in adolescents (Politte-Corn et al., 2023).

Conclusions

The current study adds to existing literature on the relation between social media use and emotional health in adolescents and lays the groundwork for future work to expand on several questions. Findings indicated that social media use is associated with reduced PA in adolescents. At the same time, social media use was associated with reduced NA, but only for currently depressed youth. These findings provide new insight into the time-linked connections between social media use and affect and how this differs for youth with clinical depression. Delineating how, when, and for whom social media use may confer risk or benefits for affective health is a key direction for future research.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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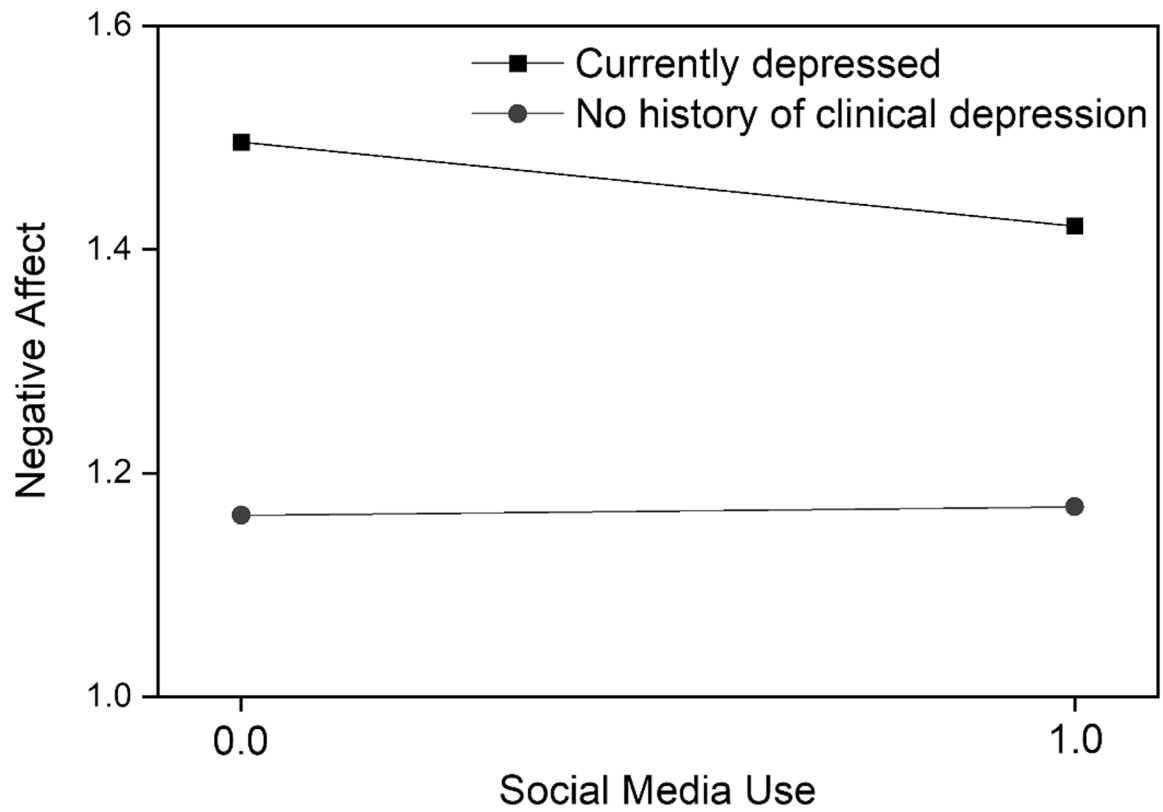


Figure 1.

2-way interaction between social media use and teen depressive disorder predicting NA at the following assessment.

Note: Across assessments, the mean value of NA was 1.26 ($SD = .51$). At each assessment, participants indicated whether they were using social media on a checklist (yes = 1; no = 0).

Table 1

Means, standard deviations, and correlations with confidence intervals (N = 145)

Variable	M	SD	1	2	3	4
1. Age	15.17	1.07				
2. Social media use	0.13	0.14	.05 [-.12, .21]			
3. PA	2.53	0.76	-.15 [-.31, .01]	.00 [-.17, .17]		
4. NA	1.31	0.41	.12 [-.04, .28]	-.06 [-.23, .11]	-.40 ^{**} [-.53, -.25]	
5. Number of assessments completed	27.30	12.72	.09 [-.07, .25]	-.09 [-.25, .08]	-.07 [-.23, .10]	-.10 [-.26, .06]

Note: Degrees of freedom = 143. Social media use in correlation analyses reflects the proportion of surveys in which use of social media was endorsed.

* $p < .05$.
** $p < .01$.

Table 2

Cross-sectional analyses of social media use predicting concurrent momentary PA and NA.

PA		β (SE)	t-value	p
Step 1	Social media use	-.04 (.01)	-3.42	<.001
Step 2	Social media use	-.04 (.01)	-3.40	<.001
	Sex	-.15 (.06)	-2.58	.01
	Depressive disorder	-.24 (.06)	-4.17	<.001
Step 3	SMU X Sex	.06 (.05)	1.20	.23
	SMU X Depressive disorder	.02 (.02)	1.19	.23
NA		β (SE)	t-value	p
Step 1	Social media use	-.01 (.01)	-0.92	.36
Step 2	Social media use	-.01 (.01)	-0.84	.40
	Sex	.03 (.06)	0.44	.66
	Depressive disorder	.29 (.06)	4.81	<.001
Step 3	SMU X Sex	-.09 (.06)	-1.64	.10
	SMU X Depressive disorder	-.03 (.02)	-1.79	.07

Note: $N = 3,910$ surveys nested within 145 participants. We retained a random slope for social media use. Interaction terms were tested in separate models to avoid overcontrolling. Sex and depressive disorder were retained as covariates when testing interaction terms. Biological sex was coded as 1 = male, 2 = female. Degrees of freedom are 2974 for social media use at Step 1 and 142 for between-person variables.

Table 3

Lagged analyses of social media use predicting momentary PA and NA at the next assessment.

PA		β (SE)	t-value	p
Step 1	Social media use	-.03 (.01)	-2.60	.009
Step 2	Social media use	-.03 (.01)	-2.63	.009
	Sex	-.15 (.06)	-2.52	.01
	Depressive disorder	-.25 (.06)	-4.06	<.001
Step 3	SMU X Sex	.06 (.05)	1.14	.25
	SMU X Depressive disorder	.02 (.02)	0.92	.36
NA		β (SE)	t-value	p
Step 1	Social media use	-.02 (.01)	-1.49	.14
Step 2	Social media use	-.02 (.01)	-1.47	.14
	Sex	.01 (.06)	0.19	.85
	Depressive disorder	.28 (.06)	4.76	<.001
Step 3	SMU X Sex	.02 (.05)	0.35	.73
	SMU X Depressive disorder	-.04 (.02)	-2.39	.02

Note: $N = 2,953$ surveys nested within 138 participants. We reverted to a fixed slope for social media use. Interaction terms were tested in separate models to avoid overcontrolling. Sex and depressive disorder were retained as covariates when testing interaction terms. Biological sex was coded as 1 = male, 2 = female. Degrees of freedom are 2158 for social media use at Step 1 and 135 for between-person variables.